Find all pairs of polar coordinates that describe the same point as the provided polar coordinates.

5) $(4, 90°)$

6) $\left(2, \dfrac{11\pi}{12}\right)$

Convert each pair of polar coordinates to rectangular coordinates.

7) $\left(2, \dfrac{3\pi}{2}\right)$

8) $\left(1, \dfrac{5\pi}{6}\right)$

Convert each pair of rectangular coordinates to polar coordinates where $r > 0$ and $0 \le \theta < 2\pi$.

9) $\left(\dfrac{3\sqrt{3}}{2}, \dfrac{3}{2}\right)$

10) $\left(-\sqrt{2}, \sqrt{2}\right)$

Two points are specified using polar coordinates. Find the distance between the points.

11) $\left(2, \dfrac{\pi}{3}\right), \left(2, \dfrac{11\pi}{6}\right)$

12) $\left(4, \dfrac{7\pi}{12}\right), \left(2, \dfrac{\pi}{12}\right)$

Critical thinking question:

13) An air traffic controller's radar display uses polar coordinates. A passing plane is detected at 285° counter-clockwise from north at a distance of 3 miles from the radar. Thirty seconds later the plane is detected at 225° and 2 miles. Estimate the plane's speed in miles per hour.